

Archisman Panigrahi

Graduate Student (Ph.D. Candidate) · Physics

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Education

Ph.D. in Physics (ongoing)

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

• C.G.P.A - 5.0/5.0

Supervisor: Prof. Leonid Levitov

Cambridge, MA, USA

August 2022 - Ongoing

Master of Science in Physics

INDIAN INSTITUTE OF SCIENCE

• C.G.P.A - 9.8/10.0

Bangalore, India

Aug. 2021 - Jun. 2022

Bachelor of Science (Research) in Physics

INDIAN INSTITUTE OF SCIENCE

• C.G.P.A - 9.8/10.0

Bangalore, India

Aug. 2017 - Jun. 2021

Research Articles

LISTED IN THE CHRONOLOGICAL ORDER OF THEIR APPEARANCE ON ARXIV.

- **A. Panigrahi**, V. Poliakov, L. Levitov; *Particle-Hole Ghost Interference in Superconductors* [arXiv:2606.06437](#)
- Y. Liao, A. Grankin, **A. Panigrahi**, V. Galitski, L. Levitov; *Tunable viscosity across the BCS-BEC crossover* [arXiv:2604.10759](#)
- **A. Panigrahi**, D. Kaplan; *Valley polarization of chiral excitonic bound states induced by band geometry* [arXiv:2604.05222](#)
- P. Zhu, **A. Panigrahi**, L. Levitov, N. Trivedi; *Strain Response as a Probe of Spinons in Quantum Spin Liquids* [arXiv:2512.03137](#)
- **A. Panigrahi**, B. Roy; *Projected branes as platforms for crystalline, superconducting, and higher-order topological phases* [PRB 113, 035301 \(2026\)](#)
- **A. Panigrahi**, V. Poliakov, L. Levitov; *Tomographic imaging of superconducting order using particle-hole interference* [PNAS 123 \(27\) e2534730123 \(2026\)](#)
- **A. Panigrahi**, A. Kumar; *Non-Fermi Liquids from Subsystem Symmetry Breaking in van der Waals Multilayers* [PRL 134, 236502 \(2025\)](#)
- **A. Panigrahi**, V. Poliakov, Z. Dong, L. Levitov; *Spin chirality and fermion stirring in topological bands* [arxiv:2407.17433](#)
- L. Holleis, T. Xie, S. Xu, H. Zhou, C. L. Patterson, **A. Panigrahi**, T. Taniguchi, K. Watanabe, L. S. Levitov, C. Jin, E. Berg, A. F. Young; *Fluctuating magnetism and Pomeranchuk effect in multilayer graphene* [Nature 640, 355–360 \(2025\)](#)
- M. Masseroni, M. Gull, **A. Panigrahi**, N. Jacobsen, F. Fischer, C. Tong, J. D. Gerber, M. Niese, T. Taniguchi, K. Watanabe, L. Levitov, T. Ihn, K. Ensslin, H. Duprez; *Spin-orbit proximity in MoS₂/bilayer graphene heterostructures* [Nat Commun 15, 9251 \(2024\)](#)
- **A. Panigrahi**, L. Levitov; *Signatures of electronic ordering in transport in graphene flat bands* [PRB 110, 035122 \(2024\)](#)
- **A. Panigrahi**, V. Juričić, B. Roy; *Projected Topological Branes* [Commun Phys 5, 230 \(2022\)](#)
- **A. Panigrahi**, S. Mukerjee; *Energy magnetization and transport in systems with a non-zero Berry curvature in a magnetic field* [SciPost Phys. Core 6, 052 \(2023\)](#)
- **A. Panigrahi**, R. Moessner, B. Roy; *Non-Hermitian dislocation modes: Stability and melting across exceptional points* [PRB 106, L041302 \(2022\)](#)

Research Experience

Aspects of spin chirality and tunneling density of states in systems with broken time-reversal symmetry

MIT, Cambridge, MA, USA

WITH PROF. LEONID LEVITOV

2024 — Present

- Demonstrated that tunneling density of states near two impurities may be utilized to probe superconducting order parameter
- Demonstrated that spin chirality is spontaneously generated in time-reversal symmetry broken systems without any spin-orbit coupling
- Predicted that this effect can be utilized in detecting topological phase winding

Non-Fermi liquids resulting from subsystem symmetry breaking

MIT, Cambridge, MA, USA

WITH AJESH KUMAR

2024

- Demonstrated that subsystem symmetry breaking in van der Waals heterostructures can give rise to an anisotropic non-Fermi liquid, with quasiparticle lifetime $\tau \sim \frac{1}{|\omega| \log |1/\omega|}$ and specific heat $C \sim T[\log(1/T)]^2$.

Transport in ordered phases in graphene

MIT, Cambridge, MA, USA

WITH PROF. LEONID LEVITOV

2023 — 2024

- Predicted that momentum-polarized nematic phases in biased bilayer graphene can lead to resistance decreasing with rising temperature
- Demonstrated hysteresis-like switching behavior under the action of a strong electric field

Many Body Localization (MBL) and thermalization of interacting quantum spin chain

IISc, Bangalore, India

(Master's thesis)

WITH PROF. SUBROTO MUKERJEE

September 2021 - April 2022

- Studied how the Out-of-Time Ordered Correlator (OTOC) behaves for MBL and thermal systems
- Studied behavior of OTOC in MBL systems with random and incommensurate potential, with and without interaction

Topological phases in projected lower dimensional branes

MPIPKS, Dresden, Germany

(remotely)

JOINTLY WITH PROF. BITAN ROY AND PROF. VLADIMIR JURIČIĆ

June 2021 - September 2021

- Verified the existence of dislocation modes, Weyl points, and Landau levels in projected crystals and Fibonacci quasicrystals
- Proposed how this method can be utilized to study higher dimensional (>3D) topological phases within 3D systems

Berry curvature effects on thermoelectric transport

IISc, Bangalore, India

(Bachelor's thesis)

WITH PROF. SUBROTO MUKERJEE

October 2020 - June 2021

- Found a condition on the energy magnetization such that the Einstein relation holds for the transport energy current in these systems
- Analytically solved the Boltzmann transport equation (including Berry curvature effects) for two-dimensional systems

Non-Hermitian Topological Insulators and Dislocations

MPIPKS, Dresden, Germany

(remotely)

WITH PROF. BITAN ROY

May 2020 - September 2020

- Obtained phase diagrams for regimes where topological states get pinned at dislocation centers
- Proposed how dislocations can be used to probe topological phases in non-Hermitian systems, where the non-Hermitian skin effect masks the traditional bulk-boundary correspondence

Research Interests

Broadly interested in theoretical Condensed Matter Physics

- Tunneling phenomena in superconductors with broken time-reversal symmetry
- Non-Fermi Liquids emerging due to subsystem symmetry breaking
- Spin chirality in systems with spontaneously broken time-reversal symmetry
- Electronic transport in two-dimensional systems and the effects of Berry curvature in transport
- Topological phases of matter and Quantum Phase transitions

Skills

Programming skills Julia, MATLAB/Octave, Mathematica, Python

Advanced Physics Courses Strongly Correlated Systems, Advanced Statistical Physics, Quantum Field Theory I, General Relativity

Languages Fluent in English, Bengali, Hindi

Talks

Tunneling density of states in exotic superconductors and spatial patterns of particle-hole interference

APS March Meeting, Anaheim, CA

CLICK [HERE](#) TO DOWNLOAD THE PRESENTATION

March 2025

Transport Signatures of Electronic Ordering in Graphene Flat Bands

Indian Institute of Science,

Bangalore, India

CLICK [HERE](#) TO DOWNLOAD THE PRESENTATION

January 2024

Topological phases in quasicrystals: A general principle of construction

APS March Meeting (virtually)

CLICK [HERE](#) TO DOWNLOAD THE PRESENTATION

March 2022

Dislocation as a bulk probe of non-Hermitian topology

MPIPKS, Dresden, Germany

(remotely)

CLICK [HERE](#) TO DOWNLOAD THE PRESENTATION

July 6, 2021

Teaching Experience

Theory of Solids II (8.512)

TEACHING ASSISTANT

- Conducted recitation classes on various topics in condensed matter physics

MIT

Spring 2026

Physics II: Electricity and Magnetism (8.102)

TEACHING ASSISTANT

- Taught students one-on-one in office hours and graded exams

MIT

Spring 2024

Academic Achievements

2023	Qualified among the top 16 participants in MIT Integration Bee	MIT
2022	1st Rank in India in CSIR-NET (JRF) 2021 in Physics, held in February 2022 due to COVID (score 186/200)	India
2022	1st Rank in India in Graduate Aptitude Test in Engineering (G.A.T.E.) in Physics	India
2017-22	CGPA 9.8/10 in B.S. (Research) and M.S., received Prof. R. Srinivasan Medal for highest CGPA in batch	IISc, Bangalore
2017	1st rank (99.2 %) in Board in Higher Secondary Examination, among about 0.7 million candidates	West Bengal, India
2015	2nd rank (97.57 %) in Board in Secondary Examination, among about 1 million candidates	West Bengal, India

References

- Prof. **Leonid Levitov**, Dept. of Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA.
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- Prof. **Subroto Mukerjee**, Dept. of Physics, Indian Institute of Science, Bangalore, India.
Email Address - smukerjee@iisc.ac.in
- Prof. **Bitan Roy**, Dept. of Physics, Lehigh University, Bethlehem, PA 18015, USA.
Email Address - bitan.roy@lehigh.edu